

ER Diagram - International Airport Database

1. System Description

The Airport Database manages airport operations, including passenger bookings, flights, airlines, and booking details. The system tracks airports, the flights that depart from and arrive at them, the airlines that operate those flights, passengers and the bookings they make, any changes made to a booking, boarding passes issued for a booking, baggage registered for a booking, baggage checks performed on registered baggage, and security checks performed on passengers.

2. Identified Entities

- Airport
- Airline
- Flight
- Passenger
- Booking
- Booking_Change
- Boarding_Pass
- Baggage
- Baggage_Check
- Security_Check

3. Entities and Attributes

3.1 Airport

Attribute	Data type	Key
airport_id	INT	PK
airport_name	VARCHAR(150)	NOT NULL
country	VARCHAR(100)	NOT NULL
state	VARCHAR(100)	-
city	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.2 Airline

Attribute	Data type	Key
airline_id	INT	PK
airline_code	VARCHAR(10)	UNIQUE, NOT NULL
name	VARCHAR(150)	NOT NULL
country	VARCHAR(100)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.3 Flight

Attribute	Data type	Key
flight_id	INT	PK
airline_id	INT	FK -> Airline
departure_airport_id	INT	FK -> Airport
arrival_airport_id	INT	FK -> Airport
departing_gate	VARCHAR(10)	-
arriving_gate	VARCHAR(10)	-
scheduled_departure_time	DATETIME	NOT NULL
scheduled_arrival_time	DATETIME	NOT NULL
actual_departure_time	DATETIME	-
actual_arrival_time	DATETIME	-
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.4 Passenger

Attribute	Data type	Key
passenger_id	INT	PK
first_name	VARCHAR(100)	NOT NULL
last_name	VARCHAR(100)	NOT NULL
gender	VARCHAR(20)	-
date_of_birth	DATE	NOT NULL
country_of_citizenship	VARCHAR(100)	NOT NULL
country_of_residence	VARCHAR(100)	NOT NULL
passport_number	VARCHAR(30)	UNIQUE, NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.5 Booking

Attribute	Data type	Key
booking_id	INT	PK
flight_id	INT	FK -> Flight
passenger_id	INT	FK -> Passenger
status	VARCHAR(30)	NOT NULL
booking_platform	VARCHAR(50)	NOT NULL
ticket_price	DECIMAL(10,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.6 Booking_Change

Attribute	Data type	Key
change_id	INT	PK
booking_id	INT	FK -> Booking
change_description	VARCHAR(255)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.7 Boarding_Pass

Attribute	Data type	Key
boarding_pass_id	INT	PK
booking_id	INT	FK -> Booking
seat	VARCHAR(5)	NOT NULL
boarding_time	DATETIME	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.8 Baggage

Attribute	Data type	Key
baggage_id	INT	PK
booking_id	INT	FK -> Booking
weight_kg	DECIMAL(5,2)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.9 Baggage_Check

Attribute	Data type	Key
baggage_check_id	INT	PK
booking_id	INT	FK -> Booking
passenger_id	INT	FK -> Passenger
check_result	VARCHAR(30)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

3.10 Security_Check

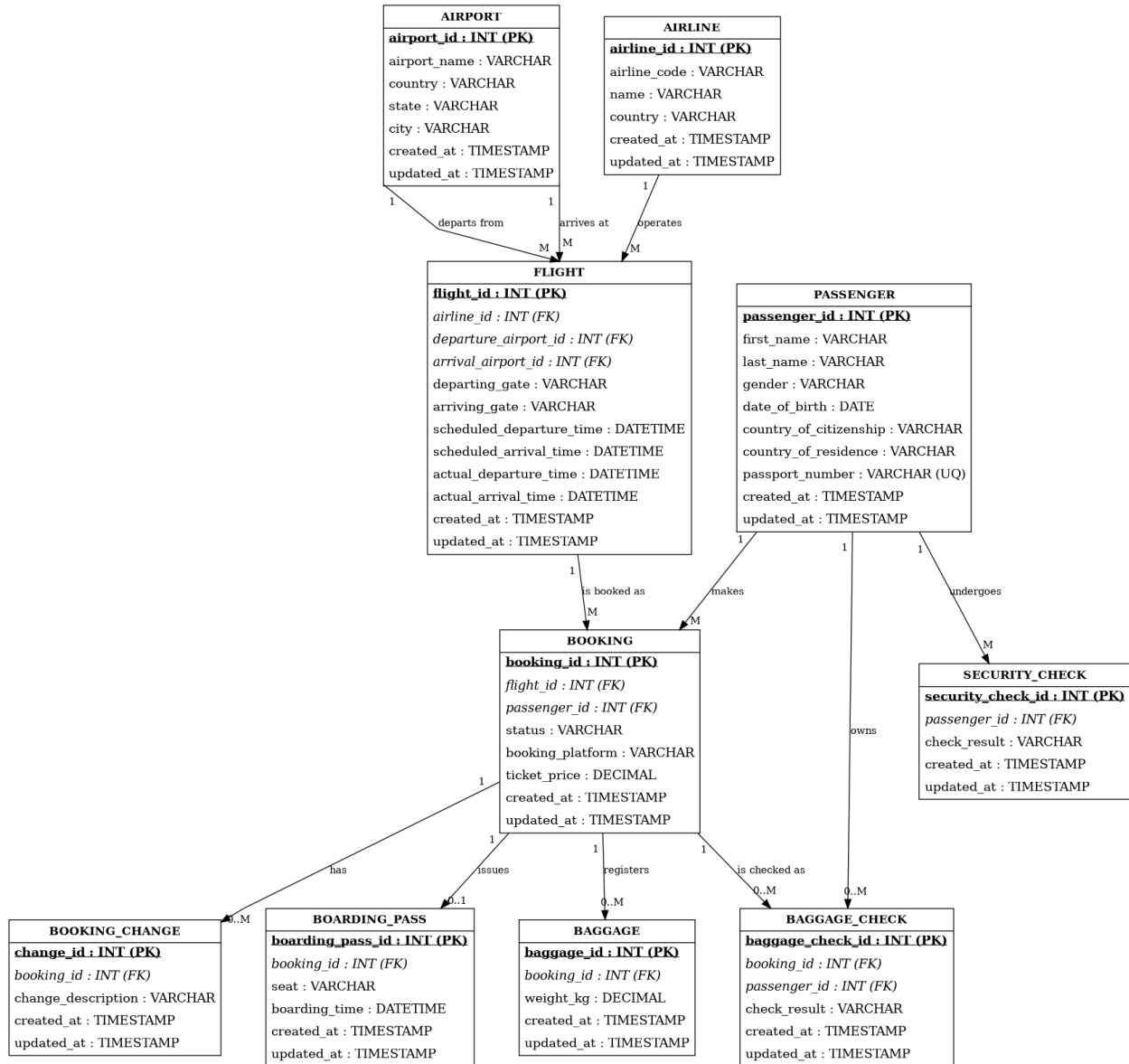
Attribute	Data type	Key
security_check_id	INT	PK
passenger_id	INT	FK -> Passenger
check_result	VARCHAR(30)	NOT NULL
created_at	TIMESTAMP	NOT NULL
updated_at	TIMESTAMP	NOT NULL

4. Identified Relationships

#	Entities	Cardinality	Participation	Description
1	Airport - Flight (departure)	1 : M	Airport optional, Flight mandatory	One airport is the departure point of many flights
2	Airport - Flight (arrival)	1 : M	Airport optional, Flight mandatory	One airport is the arrival point of many flights
3	Airline - Flight	1 : M	Airline optional, Flight mandatory	One airline operates many flights
4	Flight - Booking	1 : M	Flight optional, Booking mandatory	One flight can have many bookings
5	Passenger - Booking	1 : M	Passenger optional, Booking mandatory	One passenger can make many bookings
6	Booking - Booking_Change	1 : 0..M	Booking optional (0 or more changes)	A booking may have zero or more recorded changes
7	Booking - Boarding_Pass	1 : 0..1	Booking optional	A booking results in at most one boarding pass
8	Booking - Baggage	1 : 0..M	Booking optional (baggage is not required)	A booking may include zero or more registered baggage items
9	Booking - Baggage_Check	1 : 0..M	Booking optional	Baggage belonging to a booking is checked; results are logged per booking
10	Passenger - Baggage_Check	1 : 0..M	Passenger optional	A passenger is linked to the baggage check records of their baggage
11	Passenger - Security_Check	1 : M	Passenger mandatory	Each passenger undergoes one or more security checks

5. ER Diagram

Entity-Relationship Diagram - International Airport Database



6. Legend

- Each box represents an entity (a table in the database). The top row shows the entity name.
- Bold, underlined attribute = Primary Key (PK) - uniquely identifies each row of the entity.
- Italic attribute = Foreign Key (FK) - references the primary key of another entity, implementing the relationship.
- Plain attribute = regular column (UQ = unique constraint, NOT NULL where required by the system description).
- Lines between entities represent relationships; the label on the line names the relationship.
- Numbers next to the line ends show cardinality: 1 = exactly one, M = many, 0..1 = zero or one (optional), 0..M = zero or many (optional).
- created_at / updated_at columns appear in every entity and record when a row was created and last modified, as required by the system description.

Note: Baggage_Check stores both booking_id and passenger_id, following the checking process described in the system specification (baggage belonging to a booking is checked, and the check result is also associated with the passenger who owns that baggage).